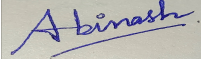
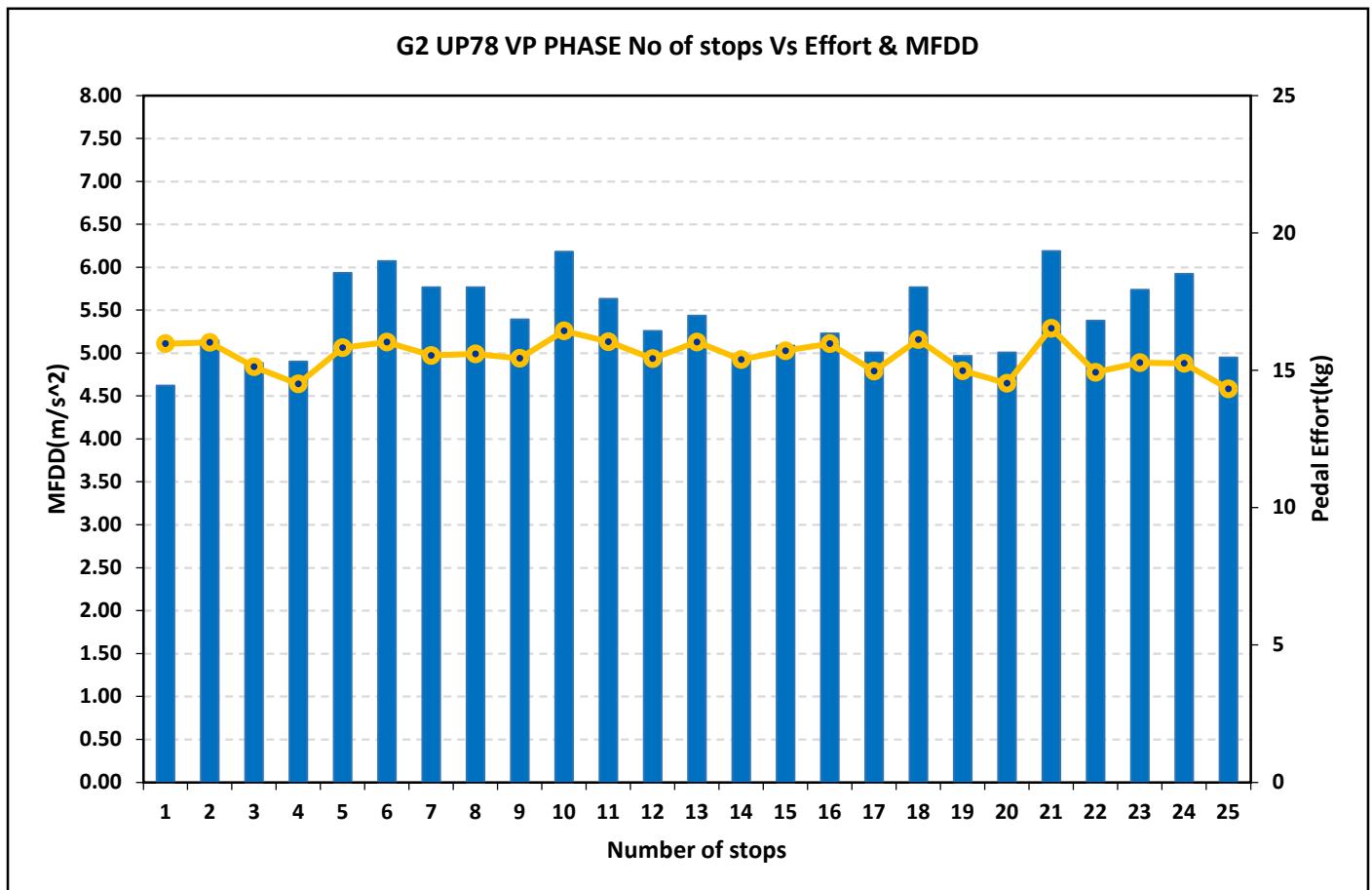
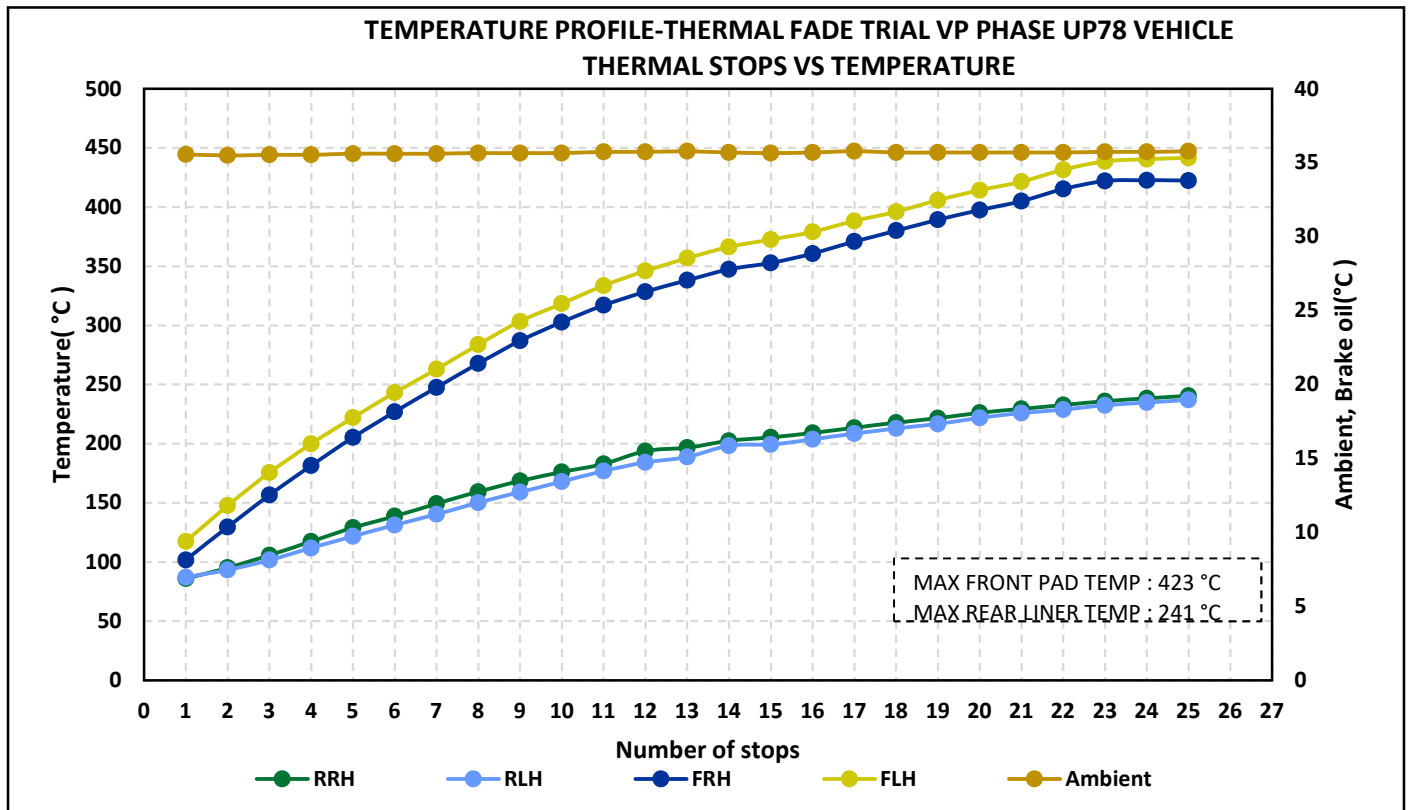
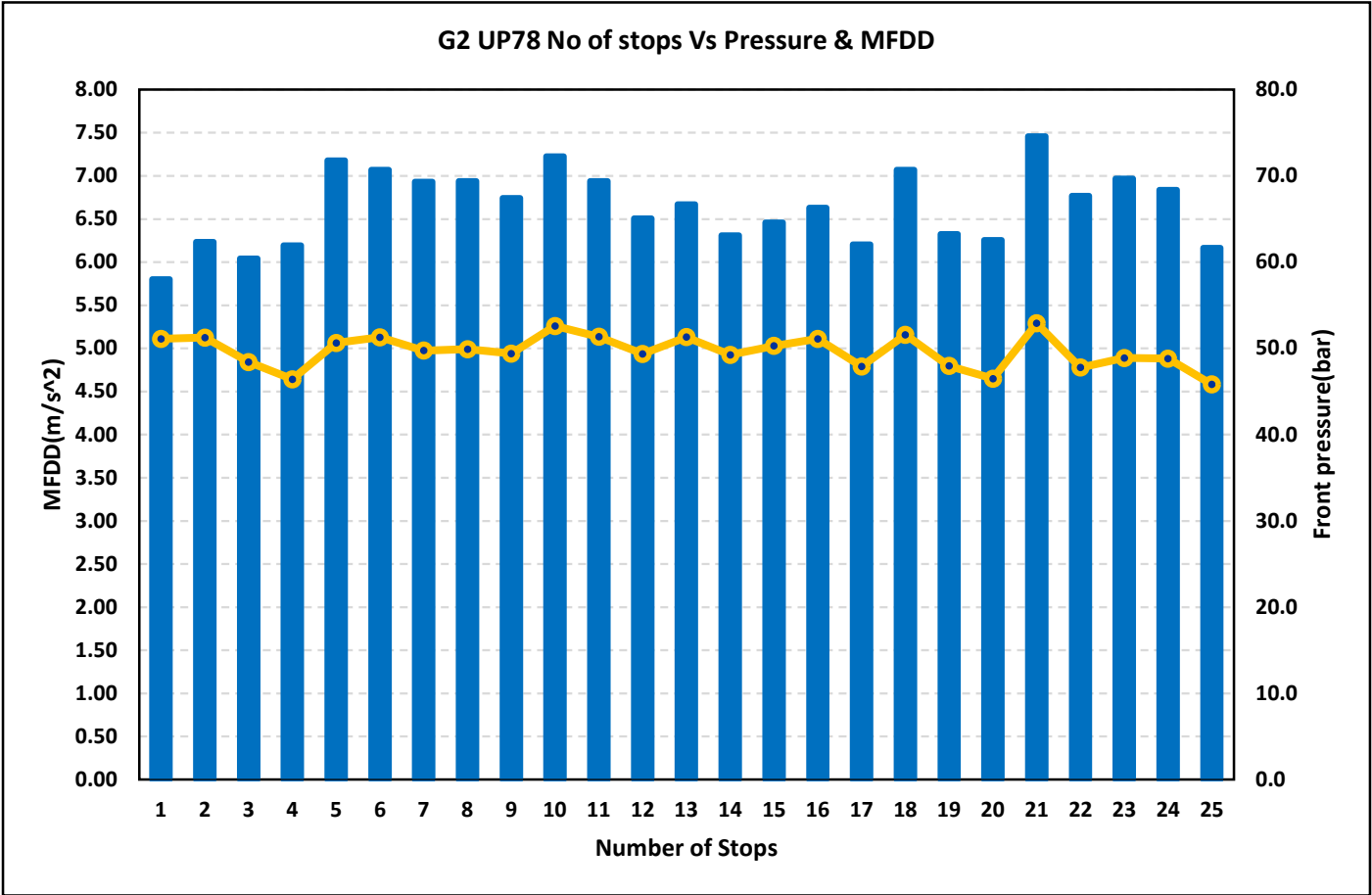


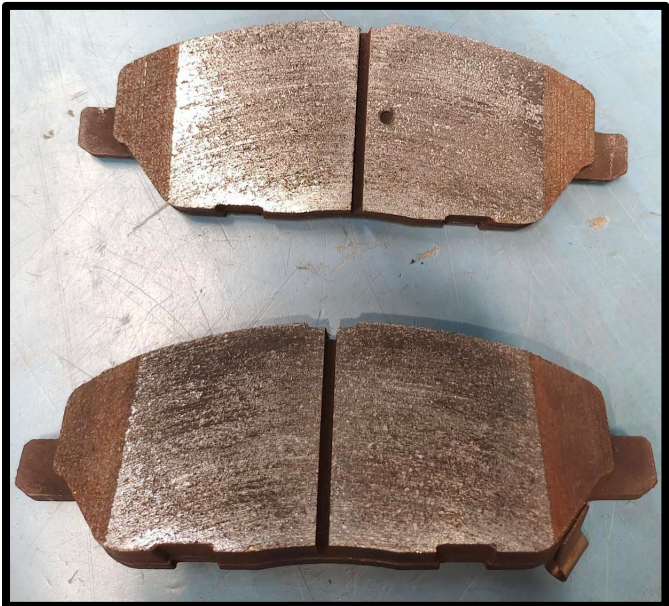
TPDS	TEST REPORT								
BRAKE THERMAL CAPACITY EVALUATION ON P201 VP PHASE UP78 VEHICLE IN GVW CONDITION									
To :	NAGRIKAR RAVI/MISHRA KRISHNA	Report No:	TPDS/VT/RDP201/VP/2024/27914						
Project No:	RDP201-UPP Goods-G2-Domestic	Report Date:	30-Apr-2024						
Objective:	To evaluate brake thermal capacity evaluation test on P201 VP phase vehicle with respect to test target.								
Scope:	To conduct brake thermal capacity evaluation test on P201 VP phase vehicle in GVW condition								
Vehicle Configuration:									
Foundation Brakes	BIL	Booster Size	9"						
Front Brake	C60, Single pot	Booster Ratio	5						
Front Disc Size, Eff radius	277 x 24, 116	Jump in	12 +/-3						
Front Brake Pad Grade	Apache 5409M	Pedal Ratio	3.5:1						
Rear Brake	25.4 mm (WC)	TMC Size	23.81						
Rear Drum size, Eff. radius	279 X 50.8, 279	Stroke-Primary/Secondary	17.5/18.5mm						
Rear Brake Pad Grade	RD0014	System Split	"X" Split						
Test:	06-C-06-0295-001-BRAKE THERMAL CAPACITY EVALUATION								
Observation:	Refer the table below for test observations:-								
<table><tr><th>TEST TARGET</th><th>OBSERVATION</th><th>REMARKS</th></tr><tr><td>1.The percentage recovery must be more than 90% with in the third 2 min cooling stop or by first 4 min cooling stop. 2.The maximum lining (pads in case of disc brakes or Liner in case of drum brakes) temperatures achieved in the fade stops must be with in the design acceptance of maximum operating temperature of the lining, and must be well within the maximum allowable temperatures of the friction bonding material. (If target is provided) 3.No friction material crack, Ash formation, burnt marks must be present on the friction material. 4.No thermal cracks and decolorization, glazing marks on the disc / drum surface. 5.No damage on the wheel end components like wheel rims, bolts, bearings, calliper seals & boot, wheel bearing grease seepage, etc.</td><td>1. 97% Recovery is achieved after 1st stop of 1 minute cooling cycle. 2. 100% recovery is achieved after 1st stop of 2-minute cooling cycle. 3. Peak temperatures recorded are 442 °C and 241°C for front pad and rear pad respectively. 4. Ash formation, burnt marks present on the brake pad friction material and brake dust formed on the wheel rim. 5. Decolorization marks on the front brake pad & disc. 6. No damage on the wheel end components like wheel rims, bolts, bearings, calliper seals & boot, wheel bearing grease seepage, etc. 7. Overall test vehicle is meeting the test requirement wrt recovery performance. 8.Squeal noise observed during braking stops in hot condition.</td><td>Test vehicle is meeting the test requirement.</td></tr></table>				TEST TARGET	OBSERVATION	REMARKS	1.The percentage recovery must be more than 90% with in the third 2 min cooling stop or by first 4 min cooling stop. 2.The maximum lining (pads in case of disc brakes or Liner in case of drum brakes) temperatures achieved in the fade stops must be with in the design acceptance of maximum operating temperature of the lining, and must be well within the maximum allowable temperatures of the friction bonding material. (If target is provided) 3.No friction material crack, Ash formation, burnt marks must be present on the friction material. 4.No thermal cracks and decolorization, glazing marks on the disc / drum surface. 5.No damage on the wheel end components like wheel rims, bolts, bearings, calliper seals & boot, wheel bearing grease seepage, etc.	1. 97% Recovery is achieved after 1st stop of 1 minute cooling cycle. 2. 100% recovery is achieved after 1st stop of 2-minute cooling cycle. 3. Peak temperatures recorded are 442 °C and 241°C for front pad and rear pad respectively. 4. Ash formation, burnt marks present on the brake pad friction material and brake dust formed on the wheel rim. 5. Decolorization marks on the front brake pad & disc. 6. No damage on the wheel end components like wheel rims, bolts, bearings, calliper seals & boot, wheel bearing grease seepage, etc. 7. Overall test vehicle is meeting the test requirement wrt recovery performance. 8.Squeal noise observed during braking stops in hot condition.	Test vehicle is meeting the test requirement.
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Conclusion:	Brake thermal capacity evaluation test on P201 VP phase vehicle is meeting the test target.								
Recommendation:	Nil								
Attachments:	Data, Graph								
Test Conducted / Reported by  Abinash Abinash		Reviewed by  MITTAL SAPAN	Approved by  Rathakrishnan M						
CC SHABBIR KARNURE/LALASURE SANTOSH									

TPDS		VEHICLE CONFIGURATION			
Project No : RDP201-UPP Goods-G2-Domestic			Report No : TPDS/VT/RDP201/VP/2024/27914		
Vehicle Details					
Parameter		P201 VP UP78 DMT Vehicle			
Model		G2 3D15 2550 195R15 40L FSD			
Reg No / S No		R6B15023			
Engine Model & No		1.5 lit, 3 Cyl CUR6B44576			
C.G.Height(MM)		902 mm			
Transmission /Gearbox		5 speed			
Drive		2WD			
Axle Ratio		--			
Wheel / Tyre Details		Tubeless 195 R15 37 PSI and 58 PSI CEAT			
Front Suspension		Rigid-Semi Elliptical (5 Leaf)			
Rear Suspension		Rigid-Semi Elliptical (2 stage -8			
Steering Details		RCBT			
Speedo Make/Raio		--			
Odometer					
Wheel Track(mm)		Front - 1465, Rear - 1375			
Wheel Base(mm)		2550			
F/R Ratio		--			
Dyn.Rolling Radius		--			
Weight Transfer Ratio		--			
Weight Distribution:					
UNLADEN	Curb Wt (kg) FAW (kg) RAW (kg) ETW (kg)	1631 1157 474 1631			
LADEN	GVW (kg) FAW (kg) RAW (kg) Pay Load (kg)	3038 1400 1638 1260			
OVERLOAD	GVW (kg) FAW (kg) RAW (kg) Pay Load (kg)	3800 1495 2305 3			
Amb Temperature		34 Deg C			
Wind Velocity		<3m/s			
Gust		< 5 m/s			
Rel. Humidity		--			
Weather		Sunny			
Road Surface		Tar			
Instruments Used		Link logger & L350			
Test Site		MSPT			
Test Driver		Abinash			
Test Engineer		Abinash			
Test Date		18-04-2024			

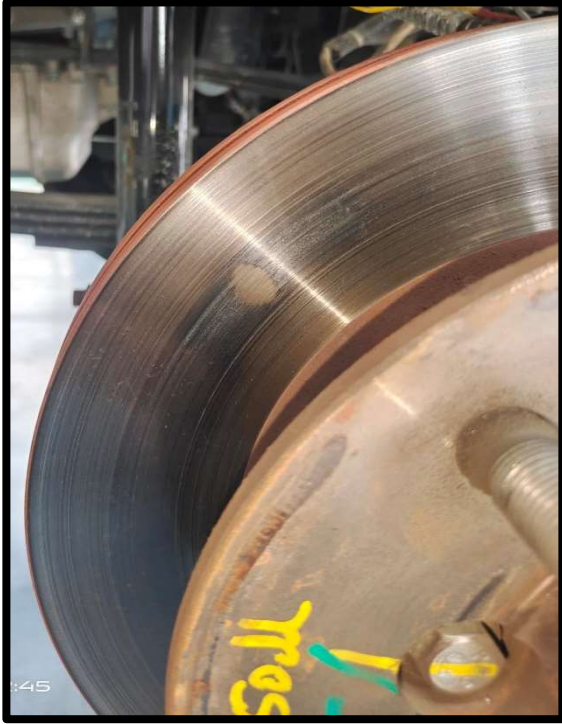




FRH PAD



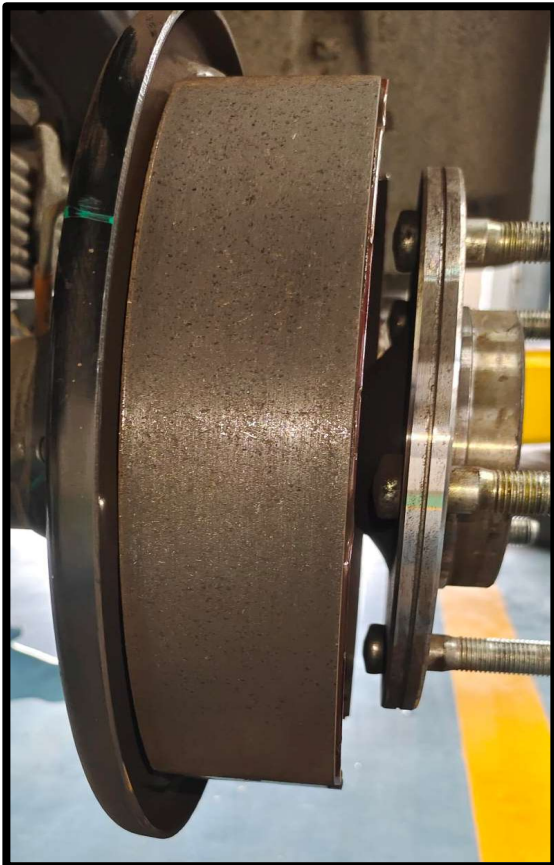
FLH PAD



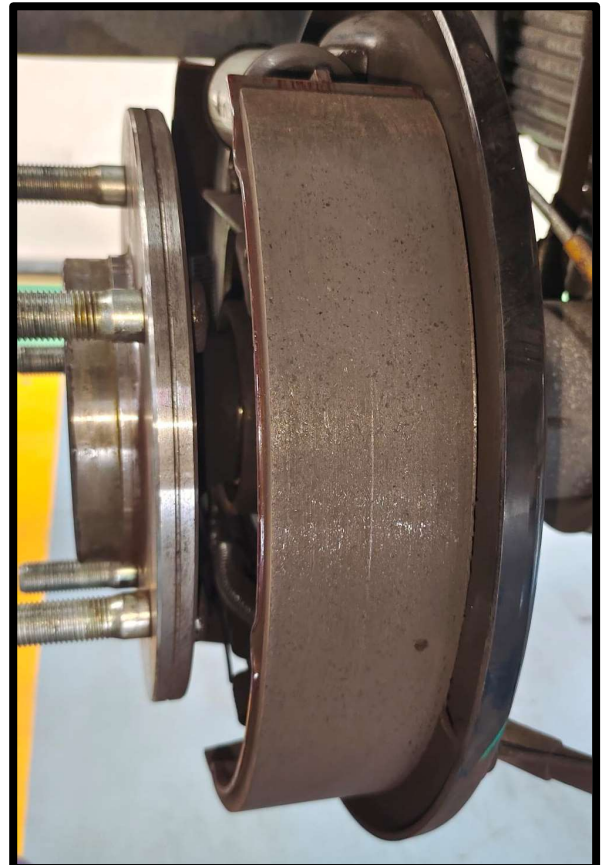
FRH DISC



FLH DISC



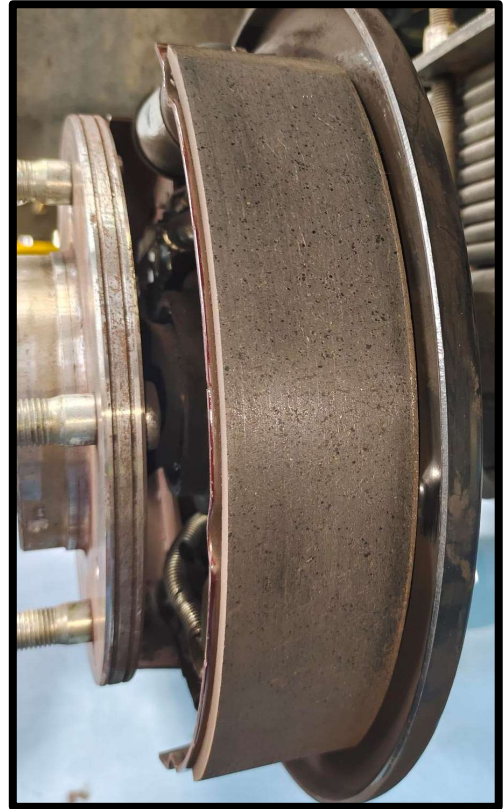
RRH LEADING



RRH TRAILING



RLH LEADING



RLH TRAILING



RRH DRUM



RLH DRUM